Joint Classification of Hyperspectral and LiDAR Data Using a Hierarchical CNN and Transformer

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***Abstract*— The joint use of multisource remote-sensing (RS) data for Earth observation missions has drawn much attention. Although the fusion of several data sources can improve the accuracy of land-cover identification, many technical obstacles, such as disparate data structures, irrelevant physical characteristics, and a lack of training data, exist. In this article, a novel dual-branch method, consisting of a hierarchical convolutional neural network (CNN) and a transformer network, is proposed for fusing multisource heterogeneous information and improving joint classification performance. First, by combining the CNN with a transformer, the proposed dual-branch network can significantly capture and learn spectral–spatial features from hyperspectral image (HSI) data and elevation features from light detection and ranging (LiDAR) data. Then, to fuse these two sets of data features, a cross-token attention (CTA) fusion encoder is designed in a specialty. The well-designed deep hierarchical architecture takes full advantage of the powerful spatial context information extraction ability of the CNN and the strong long-range dependency modeling ability of the transformer network based on the self-attention (SA) mechanism. Four standard datasets are used in experiments to verify the effectiveness of the approach. The experimental results reveal that the proposed framework can perform noticeably better than state-of-the-art methods. The source code of the proposed method will be available publicly at https://github.com/zgr6010/Fusion_HCT.git.**

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***Index Terms*— Convolutional neural network (CNN), hyperspectral image (HSI), joint classification, light detection and ranging (LiDAR) data, tokenization, transformer.**

I. Introduction

REMOTE-sensing (RS) imaging technology has improved significantly in recent years, increasing the availability of RS images [1]. Often, data from several kinds of RS devices for the same geographic region are available, making it possible to use multimodal data to analyze land-cover information. Multiple different sensor technologies can efficiently capture different characteristics of land-cover materials. For example, hyperspectral image (HSI) sensors can obtain spectral information along with geographical spatial information [2], while light detection and ranging (LiDAR) or digital surface model (DSM) measures the elevation and object height information of the Earth's surface [3], and synthetic aperture radar provides geometric information on the amplitude and phase of ground objects [4]. By integrating the data from these different modalities, we can obtain more detailed information, enabling us to construct an entire feature representation [5], [6]. This approach has aided in the interpretation of RS imagery in a wide assortment of research and application fields, including land-cover classification [7], [8], [9], target detection [10], [11], semantic segmentation [12], environmental monitoring [13], mineral exploration [14], urban planning [15], medical diagnosis [16], precision agriculture [17], and food security [18]. Land-cover classification has long been a particularly prominent application [19], [20].

Early work on classifying RS data mostly relied on a single kind of data. For instance, HSI classification specifically only uses HSI data. Current HSI classification techniques can achieve satisfactory results, however, some misclassification may occur, particularly in urban environments, where roads and roofs of buildings may be constructed of the same material and have similar spectral curves. Therefore, it is challenging to distinguish these land-cover types using only spectral data. Fortunately, LiDAR-based DSM data can determine the elevation of these two types of objects, making it straightforward to distinguish them. If the data from these two modalities can be combined effectively, they can compensate for each other, thereby improving classification accuracy.

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Based on classical machine-learning algorithms, numerous traditional RS land-cover classification methods have been proposed, such as support vector machines (SVMs) [21], [22], random forests (RFs) [23], and extreme learning machine [24]. Such methods rely on a single HSI and use only spectral information, resulting in limited classification performance. Data dimensionality reduction and manual feature extraction methods, used in the data preprocessing stage, offer a fresh strategy to the classification community. Principal component analysis (PCA) has a solid reputation for highly effective dimensionality reduction in HSI classification [25], [26]. This reduces the required computational power, but because only spectral dimension features are considered, it still cannot achieve satisfactory accuracy. To extract the spatial information in HSIs as features for classification, scholars have applied image spatial feature extraction methods, including morphological profiles [27], attribute profiles [28], and extinction profiles [29]. By applying a graph-based fusion strategy, Liao et al. [30] improved the extraction of morphological profiles for HSIs and LiDAR data feature fusion and classification. To further improve the joint classification accuracy, Rasti et al. [31] combined joint feature extraction of extinction profiles with total variation component analysis for feature fusion. Hong et al. [32] proposed to construct a specific feature space to share multimodal data feature information by using local preserving projection and linear graph embedding. However, these methods use shallow feature classification models that struggle to handle complex sample data distributions and highly nonlinear data. Furthermore, as these methods heavily rely on prior information, it is challenging to increase classification accuracy while preserving algorithm robustness.

The rapid advancement of deep learning has fostered innovation in many areas of image processing technology [33], [34], including in the processing of RS images. For HSI classification tasks, many deep neural network models have been proposed. The convolutional neural network (CNN), which is based on the unified convolution operation, is particularly effective in extracting features from 2-D image data. The CNN has, therefore, become a popular approach to HSI classification. Hu et al. [35] first proposed a classification method that made use of a 1-D CNN with five convolutional layers. Using the spectral information as input, it successfully extracted spectral features but did not take the spatial information into account. To address this shortcoming, Zhao and Du [36] proposed a 2-D CNN model for classifying HSIs, which extracted image spatial features from the data after PCA dimensionality reduction. Nevertheless, both the above methods only considered features in one feature dimension. Hence, Yang et al. [37] integrated 1-D and 2-D CNNs to design a dual-branch network structure to extract spectral and spatial features. Meanwhile, Chen et al. [38] developed a novel 3-D CNN technique for successfully extracting joint spatial–spectral information from HSI data. To compensate for the information loss, He et al. [39] proposed a residual network that used residual links after each convolution operation. Following that, Zhong et al. [40] created a spatial–spectral joint residual network that can effectively use spatial–spectral information to improve classification performance. To take advantage of the residual behavior of the network, Paoletti et al. [41] designed a pyramid residual network to classify HSIs. This framework successfully extracts the feature information uncovered by the convolution filter by increasing the dimension of the feature map and arranging the features into pyramid residual blocks.

Algorithms based on deep learning have also shown great potential in the joint classification of HSI and LiDAR data. To extract compact features from HSI and LiDAR data, Hong et al. [42] developed a deep encoder and decoder network to classify multimodal RS data on the pixel level. To further improve joint classification accuracy, Xu et al. [43] developed a novel dual-branch deep CNN architecture for patch-wise classification and fusion of multisource RS data. The spatial and spectral information of the HSI data were extracted using a dual-tunnel CNN branch, while the spatial information of the other RS data was extracted using a cascade network block. However, the subsequent feature fusion of this model is somewhat poor. To optimize the feature fusion stage of HSI and LiDAR data, Hang et al. [44] proposed a coupled CNN network and designed simultaneous feature-level fusion and decision-level fusion strategies to further enhance the classification performance. Meanwhile, Wu et al. [45] proposed a neural network framework that used the CNN as the backbone, with a cross-channel reconstruction module as a feature fusion stage to further enhance the feature fusion of multimodal RS data. Zhang et al. [46] proposed a novel interleaving perception CNN to integrate heterogeneous information and improve joint classification accuracy. More recently, Hong et al. [47] proposed a unified multimodal deep-learning framework focusing on RS data classification and extensively researched and summarized the multistage and multimodal fusion strategies, thus providing a comprehensive guide for multimodal feature learning and cross-modal data processing.

All the methods mentioned above are based on the CNN and its derivatives. Even though these methods significantly improve the classification performance of HSI and LiDAR data, they are often challenging to implement because of problems such as limited training data and high computational costs due to excessive feature redundancy. Additionally, while the CNN is good at representing spatial features, it is unable to handle the voluminous spectral sequence information contained in HSI data, making it difficult to discern minor spectral differences between pixels. Recurrent neural networks (RNNs) can build sequential models to effectively extract spectral features from long-range spectral bands. Following that, combining the CNN with the RNN, Wu et al. [48] designed a convolutional RNN framework. First, the former spatial–spectral features were extracted using the convolution layers. Then, the contextual spectral features from the former features were further extracted using the recurrent layers. However, learning long-distance correlation and avoiding gradient vanishing problems, which could make it more difficult to perceive spectrally salient changes between pixels, depend heavily on the sequence of the spectral bands. This is because HSIs comprise many samples, and RNN cannot train the multi-sample network simultaneously, which limits the classification performance.

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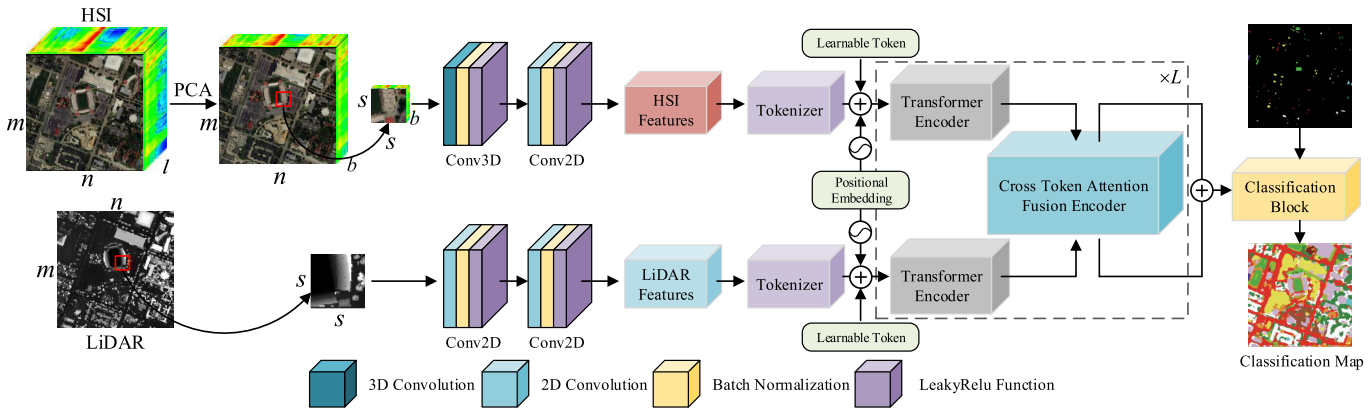


Fig. 1. Overall architecture of the proposed network for the HSI and LiDAR fusion and classification.

Transformer gradually came to the attention of researchers because of its capacity to model long sequence correlations. It was initially proposed for use in natural language processing and has since been applied to image processing [49], with some work applying it to RS classification tasks. By integrating a spectral attention mechanism with the multihead attention mechanism in the transformer, Qing et al. [50] were successful in capturing the sequential spectral relationship to improve HSI classification performance. Hong et al. [51] created a novel spectral transformer model, which can build a cross-layer transformer encoder (TE) module and capture spectral features from group-wise nearby bands. However, these works only employed transformer techniques for spectral information processing. Although the transformer succeeds at capturing spectral features, it cannot utilize spatial information. Therefore, Sun et al. [52] proposed a spectral–spatial feature tokenization transformer (SSFTT) method. This method uses a TE to model local spatial contextual information, representing neighboring sequence relations. Recently, Roy et al. [53] proposed a multimodal fusion transformer for HSI and LiDAR joint classification. This method uses LiDAR data as a learnable token to perform feature learning together with HSI tokens. This operation cannot fully fuse the effective information from the two kinds of data, thus limiting the classification accuracy.

To fully utilize the information in HSI and LiDAR data and increase classification accuracy, a novel joint classification network that incorporates a hierarchical CNN and a transformer is proposed. In this framework, a dual-branch HSI and LiDAR feature representation network and a specially designed feature fusion structure are combined to improve the joint classification accuracy. HSI hierarchical CNN and LiDAR 2-D CNN dual-branch networks are designed to extract the HSI spatial–spectral features and the LiDAR elevation features of land-cover objects synchronously. Then, a parallel feature tokenization module converts two groups of features into feature tokens. After that, two groups of feature tokens are used as inputs to the TE for feature learning. Next, a cross-token attention (CTA) fusion encoder is developed to fuse the information from two groups of feature tokens. Finally, two learnable tokens are combined to feed into a softmax-based linear layer to obtain the pixel's label.

The major contributions of this article are summarized as follows.

1) To fully extract and learn HSI spatial–spectral features and LiDAR elevation features of land-cover objects, a dual-branch hierarchical network for HSI and LiDAR data joint classification is designed by combining the CNN and transformer networks. This replaces a single CNN-based network and increases joint classification performance.
2) A CTA fusion encoder module is designed to efficaciously fuse the information from the HSI and LiDAR data. To the maximum extent possible, the HSI and LiDAR feature tokens are fused by representation learning from one another.
3) From shallow feature extraction to deep feature fusion, the systematic combination of the hierarchical CNN and transformer can adequately learn the spatial–spectral features in HSIs and elevation information in LiDAR, thereby significantly enhancing joint classification accuracy. Extensive experiments on four classical public datasets validate the effectiveness of the proposed network framework.

The remainder of this article is organized as follows. Section II describes the proposed framework. In Section III, the experimental datasets, experimental settings, and analysis of classification results are presented. The conclusions of this work and some discussions on possible future research directions are provided in Section IV.

II. Methodology

The proposed framework for joint classification of HSI and LiDAR data is illustrated in Fig. 1. The six stages that make up this framework are HSI and LiDAR data preprocessing, HSI and LiDAR feature extraction via CNNs, HSI and LiDAR feature tokenization, TE, CTA fusion encoder, and classification block.

A. HSI and LiDAR Data Preprocessing

Consider a set of HSI data, denoted by $\boldsymbol{X}_H \in \mathbb{R}^{m\times n\times l}$ and a set of LiDAR data covering the same area on the Earth's surface, denoted by $\boldsymbol{X}_L \in \mathbb{R}^{m\times n}$, where m and n

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represent the two spatial sizes and l is the number of HSI spectral bands. Each pixel can be represented by a one-hot category vector. The typically large number of spectral bands in the HSI data convey valuable information, but result in large dimensions and computationally expensive processing. To reduce the spectral dimension, PCA is used to extract the first b principal components of $\boldsymbol{X}_H$, maintaining the spatial dimensions but lowering the number of bands from l to b. Thus, after PCA dimension reduction, the HSI data $\boldsymbol{X}_H$ are transformed into $\boldsymbol{X}_H^{pca} \in \mathbb{R}^{m\times n\times b}$.

Next, for each pixel, 3-D and 2-D patch extraction are performed to obtain a small patch cube $\boldsymbol{X}_H^P \in \mathbb{R}^{s\times s\times b}$ and a small patch $\boldsymbol{X}_L^P \in \mathbb{R}^{s\times s}$, respectively, where $s \times s$ is the patch size. The indices of the center pixel are used to label each patch. For edge pixels, a padding operation on these pixels is performed—the width of padding is $(s-1)/2$. Thus, the number of small patch cubes of the HSI data and small patches of the LiDAR data are both $m \times n$. After the pixel patches with label zero are removed, the remaining sample patches are divided into a training set and a test set.

B. HSI and LiDAR Feature Extraction via CNNs

Based on their superior ability to model spatial contextual features, CNNs perform exceptionally well in HSI classification tasks. In addition, thanks to the feature extraction characteristics of 3-D CNNs, high-dimensional spectral information can be locally modeled and the depth of the output feature maps can be controlled. It has been demonstrated that CNNs are capable of obtaining the high-level features of the HSIs and are equally applicable to feature extraction from other modalities of RS data. Dual-branch CNNs are therefore used in the proposed framework to extract spatial–spectral features from the HSI data and elevation information from the LiDAR data, respectively.

As shown in Fig. 1, Conv3-D and Conv2-D are applied as two sequential layers to extract discriminative spatial and spectral features from the HSI data. Each HSI patch cube $\boldsymbol{X}_H^P$ of size $s \times s \times b$ in the training patches set is used as the input training data to the Conv3-D layer. In the Conv3-D layer, the size of convolution kernels is set to $8@3 \times 3 \times 3$. Feature cubes, produced by Conv3-D, flatten the spatial dimension into a vector as the input for the following layer. The Conv2-D layer uses $64@3\times3$ convolution kernels. After the convolution operation, 64 2-D feature maps can be generated. Unlike the HSI processing, two Conv2-D layers are used to extract the LiDAR elevation features. Each LiDAR patch $\boldsymbol{X}_L^P$ of size $s \times s$ serves as the input to the two Conv2-D layers. In the two layers, the sizes of convolution kernels are set to $16@3\times3$ and $64@3\times3$, respectively. To regularize and speed up the training process, a batch normalization layer and a rectified linear unit layer are consecutively applied after each convolutional layer.

C. HSI and LiDAR Feature Tokenization

The spatial–spectral features of HSIs and the elevation features of LiDAR extracted by the dual-branch convolutional network contain rich discriminative information and can be used for HSI classification tasks. To further embed the features to the transformer network, the extracted features require tokenization [52].

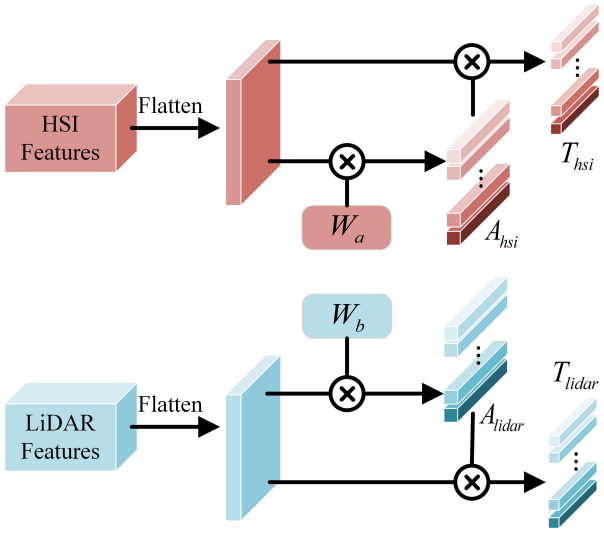


Fig. 2. Tokenization process of HSI and LiDAR features.

As shown in Fig. 2, the feature maps are flattened into a set of vectors. The flattened HSI and LiDAR feature maps are denoted by $\boldsymbol{X}_H^{\text{conv}} \in \mathbb{R}^{u_h v_h \times z_h}$ and $\boldsymbol{X}_L^{\text{conv}} \in \mathbb{R}^{u_l v_l \times z_l}$, respectively, where u_h and v_h are the height and width of the HSI feature maps, respectively, u_l and v_l are the height and width of the LiDAR feature maps, respectively, and z_h and z_l is the number of feature channels of HSI and LiDAR feature maps, respectively. The HSI feature tokens are denoted by $\boldsymbol{T}_{\text{hsi}} \in \mathbb{R}^{w_h \times z_h}$, where w_h is the number of HSI tokens. The LiDAR feature tokens are expressed as $\boldsymbol{T}_{\text{lidar}} \in \mathbb{R}^{w_l \times z_l}$, where w_l is the number of LiDAR tokens. The tokenization process uses two learnable weights, $\boldsymbol{W}_a$ and $\boldsymbol{W}_b$, which are multiplied by the input features to extract the remarkable features. For features $\boldsymbol{X}_H^{\text{conv}}$ and $\boldsymbol{X}_L^{\text{conv}}$, $\boldsymbol{T}_{\text{hsi}}$ and $\boldsymbol{T}_{\text{lidar}}$ can be obtained by the following equations:

$$\begin{aligned}\boldsymbol{T}_{\text{hsi}} &= \underbrace{\text{softmax}\left(\boldsymbol{X}_H^{\text{conv}}\boldsymbol{W}_a\right)'}_{\boldsymbol{A}_{\text{hsi}}}\boldsymbol{X}_H^{\text{conv}} \\ \boldsymbol{T}_{\text{lidar}} &= \underbrace{\text{softmax}\left(\boldsymbol{X}_L^{\text{conv}}\boldsymbol{W}_b\right)'}_{\boldsymbol{A}_{\text{lidar}}}\boldsymbol{X}_L^{\text{conv}}.\end{aligned} \tag{1}$$

Here, $\boldsymbol{X}_H^{\text{conv}}\boldsymbol{W}_a$ and $\boldsymbol{X}_L^{\text{conv}}\boldsymbol{W}_b$ are the 1×1 point-wise products. Then, the remarkable feature groups obtained by this step are transposed. The feature group after transpose are denoted by $\boldsymbol{A}_{\text{hsi}}$ and $\boldsymbol{A}_{\text{lidar}}$, respectively. Finally, $\boldsymbol{A}_{\text{hsi}}$ is multiplied by $\boldsymbol{X}_H^{\text{conv}}$ and $\boldsymbol{A}_{\text{lidar}}$ is multiplied by $\boldsymbol{X}_L^{\text{conv}}$ to yield feature tokens $\boldsymbol{T}_{\text{hsi}}$ and $\boldsymbol{T}_{\text{lidar}}$.

D. Transformer Encoder

The TE component is used to model the deep semantic relationships between feature tokens. Based on the multihead self-attention (MHSA) mechanism, the TE is able to capture the correlation information between tokens. The internal correlation created by combining multiple SA steps reduces reliance on external information.

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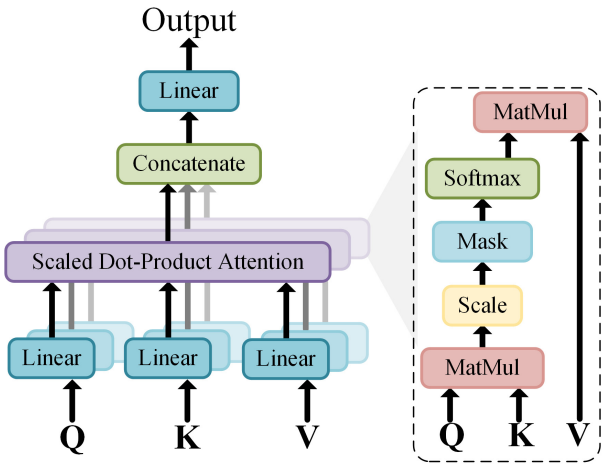


Fig. 3. Graphical illustration of the MHSA module. (Left) MHSA. (Right) SA.

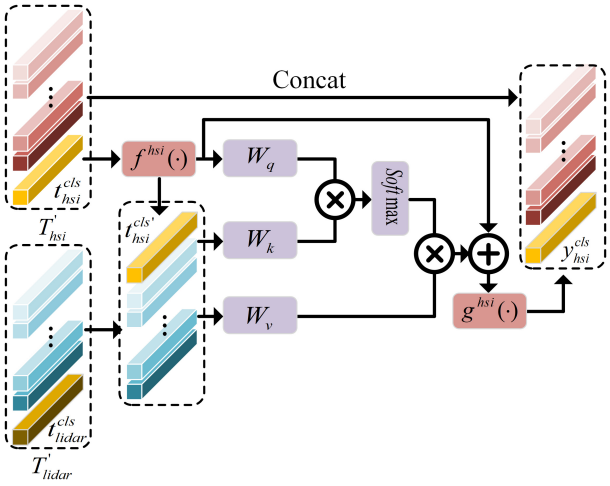


Fig. 4. Graphical illustration of the cross-token attention.

The MHSA module is illustrated in Fig. 3. SA is the core of the MHSA module. At first, the feature tokens are concatenated with an all-zero learnable classification token cls, which is used to perform the classification task. Position embedding is performed on the feature tokens to annotate position information. Then, to learn the relationships between feature tokens, three learnable weights are defined in advance for SA, including $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$. By multiplying the feature tokens with the three learnable weights, they are linearly packed together into three different matrices, namely, queries $\boldsymbol{Q}$, keys $\boldsymbol{K}$, and values $\boldsymbol{V}$. The attention scores are calculated using $\boldsymbol{Q}$ and $\boldsymbol{K}$, with the softmax function being used to translate the scores into weight probabilities. In summary, SA can be unified into a single formula

$$\mathrm{SA}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V})=\mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_{\boldsymbol{K}}}}\right)\boldsymbol{V} \tag{2}$$

where $d_{\boldsymbol{K}}$ represents the dimension of $\boldsymbol{K}$.

When mapping features to $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$, the MHSA module uses multiple groups of weights and uses the same computational process to obtain MHSA scores. All these attention scores are then concatenated together. The MHSA process is shown as follows:

$$\mathrm{MHSA}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V})=\mathrm{Concat}(\mathrm{SA}_1,\mathrm{SA}_2,\ldots,\mathrm{SA}_h)\boldsymbol{W} \tag{3}$$

where h is the number of attention heads and $\boldsymbol{W}$ is the parameter matrix, $\boldsymbol{W}\in\mathbb{R}^{h\times d_k\times d_v}$, where $d_v=w$ (number of tokens). In the TE module, the sizes of the inputs $\boldsymbol{T}_{\mathrm{hsi}}$ and $\boldsymbol{T}_{\mathrm{lidar}}$ and the outputs $\boldsymbol{T}'_{\mathrm{hsi}}$ and $\boldsymbol{T}'_{\mathrm{lidar}}$ are equal.

E. CTA Fusion Encoder

Feature fusion is essential for successfully learning multimodal feature representations for RS data since it can help to learn discriminative features by supplying additional information from other modalities of data. For this reason, a CTA fusion encoder is carefully designed to fuse the HSI and LiDAR data features [54], [55]. Specifically, to fuse two modalities of data more effectively, the cls token in the feature tokens from one modal of data is employed as an agent to exchange information among the feature tokens from another modal of data. It is then projected back to its feature tokens. Since the cls token already learns abstract information among all of its feature tokens, interacting with the feature tokens from another modal data helps to learn complementary information.

Fig. 4 shows a graphical illustration of the CTA module for the HSI feature tokens. It first concatenates its cls token with the LiDAR feature tokens, as shown in the formula below

$$\begin{aligned} t_{\mathrm{hsi}}^{\mathrm{cls}'} &= \boldsymbol{f}^{\mathrm{hsi}}\left(t_{hsi}^{\mathrm{cls}}\right)\\ \boldsymbol{T}_{\mathrm{hsi}}^{\mathrm{cta}} &= \left[\left(t_{\mathrm{hsi}}^{\mathrm{cls}'}\right)\cup\left(\boldsymbol{T}'_{\mathrm{lidar}}-t_{\mathrm{lidar}}^{\mathrm{cls}}\right)\right] \end{aligned} \tag{4}$$

where $t_{\mathrm{hsi}}^{\mathrm{cls}}$ denotes the cls token on the HSI feature tokens and has a size of $1\times z_h$. $t_{\mathrm{hsi}}^{\mathrm{cls}'}$ represents the cls token of the same dimension as the LiDAR cls token with a size of $1\times z_l$. $t_{\mathrm{lidar}}^{\mathrm{cls}}$ denotes the cls token on the LiDAR feature tokens and has a size of $1\times z_l$. $\boldsymbol{f}^{\mathrm{hsi}}(\cdot)$ is a linear projection function for dimension alignment. It uses a linear layer to transform the input $t_{\mathrm{hsi}}^{\mathrm{cls}}$ into $t_{\mathrm{hsi}}^{\mathrm{cls}'}$, which has the same dimension as $t_{\mathrm{lidar}}^{\mathrm{cls}}$. $\boldsymbol{T}_{\mathrm{hsi}}^{\mathrm{cta}}$ is expressed as the new feature tokens, where the original cls token $t_{\mathrm{lidar}}^{\mathrm{cls}}$ on the LiDAR feature tokens is replaced by the transformed HSI cls token $t_{\mathrm{hsi}}^{\mathrm{cls}'}$.

Then, the fusion encoder performs CTA between $t_{\mathrm{hsi}}^{\mathrm{cls}'}$ and $\boldsymbol{T}_{\mathrm{hsi}}^{\mathrm{cta}}$, where the cls token is the only query as the information of the HSI feature tokens are fused into the cls token. Mathematically, the CTA can be represented as

$$\begin{aligned} &\boldsymbol{Q}=t_{\mathrm{hsi}}^{\mathrm{cls}'}\boldsymbol{W}_q,\quad \boldsymbol{K}=\boldsymbol{T}_{\mathrm{hsi}}^{\mathrm{cta}}\boldsymbol{W}_k,\quad \boldsymbol{V}=\boldsymbol{T}_{\mathrm{hsi}}^{\mathrm{cta}}\boldsymbol{W}_v,\\ &\boldsymbol{A}=\mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{C/h}}\right),\quad \mathrm{CTA}\left(\boldsymbol{T}_{\mathrm{hsi}}^{\mathrm{cta}}\right)=\boldsymbol{A}\boldsymbol{V} \end{aligned} \tag{5}$$

where $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v\in\mathbb{R}^{C\times(C/h)}$ are learnable weights, C is the embedding dimension, and h is number of attention heads.

The time and space complexity of creating the attention map ($\boldsymbol{A}$) are both linear because in the query only the cls token was used, making the entire computation process more efficient. Just as in MHSA, multiple heads are used in the CTA

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module, so it can be written as MHCTA. Finally, the output $\boldsymbol{T}_{\text{hsi}}^{\text{out}}$ of the CTA of a given $\boldsymbol{T}'_{\text{hsi}}$ with layer normalization (LN) and residual shortcut is defined as

$$\begin{aligned} y_{\text{hsi}}^{\text{cls}'} &= t_{\text{hsi}}^{\text{cls}'} + \text{MHCTA}\left(\text{LN}\left(\boldsymbol{T}_{\text{hsi}}^{\text{cta}}\right)\right) \\ y_{\text{hsi}}^{\text{cls}} &= \boldsymbol{g}^{\text{hsi}}\left(y_{\text{hsi}}^{\text{cls}'}\right) \\ \boldsymbol{T}_{\text{hsi}}^{\text{out}} &= \left[y_{\text{hsi}}^{\text{cls}} \cup \left(\boldsymbol{T}'_{\text{hsi}} - t_{\text{hsi}}^{\text{cls}}\right)\right] \end{aligned} \tag{6}$$

where $y_{\text{hsi}}^{\text{cls}'}$ represents the cls token of the same dimension as the LiDAR cls token with a size of $1 \times z_l$. $y_{\text{hsi}}^{\text{cls}}$ represents the cls token of the same dimension as the HSI cls token with a size of $1 \times z_h$. $\boldsymbol{g}^{\text{hsi}}(\cdot)$ is the linear back-projection function for dimension alignment. It uses a linear layer to transform the input $y_{\text{hsi}}^{\text{cls}'}$, whose dimension is different from the cls token on the HSI feature tokens, into the same dimension $y_{\text{hsi}}^{\text{cls}}$.

The same process is performed for the LiDAR feature tokens. For LiDAR, the output is defined as $\boldsymbol{T}_{\text{lidar}}^{\text{out}}$, the learnable cls token is defined as $y_{\text{lidar}}^{\text{cls}}$.

F. Classification Block

The cls tokens $y_{\text{hsi}}^{\text{cls}}$ and $y_{\text{lidar}}^{\text{cls}}$ in the output of CTA are fed into a multi-layer perceptron (MLP) layer for the final classification, respectively. Each MLP includes two linear layers with a Gaussian error linear unit (GELU) operation, where the last linear layer with a softmax function aims to generate the final labels for joint classification. In the MLP, the size of the output layer is set to be the same as the total number of classes. Softmax ensures that the activation of each output unit sums to 1. Thus, the output can be seen as a set of conditional probabilities. The final two output probability vectors are added to get the final probability vector. The label corresponding to the maximum probability is the category of the pixel.

The cross-entropy loss function is utilized as the loss function in the integrated network, as shown in the formula below

$$H(p, q) = -\sum_{k=1}^{n} p(x_k) \log q(x_k) \tag{7}$$

where $p(x_k)$ is the predictive probability that the pixel belongs to class k. $q(x_k)$ is the expected probability that the pixel belongs to class k.

The overall training process of the proposed hierarchical CNN and transformer method is shown in Algorithm 1.

Algorithm 1 Hierarchical CNN and Transformer Fusion Network for HSI and LiDAR Data Joint Classification

Input: Input an HSI data $\boldsymbol{X}_H \in \mathbb{R}^{m \times n \times l}$, an LiDAR data $\boldsymbol{X}_L \in \mathbb{R}^{m \times n \times l}$ and a ground-truth data $\boldsymbol{Y} \in \mathbb{R}^{m \times n}$; PCA bands number $b = 30$; patch size $s = 11$; training sample rate $\phi\%$.

Output: Predicted labels of the test set.

Set batch-size to 64, optimizer Adam (learning rate: $1e-3$), epochs number e to 100, initialize all weights.
Obtain the $\boldsymbol{X}_H^{pca}$ after PCA transform.
Create all sample patches from $\boldsymbol{X}_H^{pca}$ and $\boldsymbol{X}_L$, and divide them into a training set and a test set.
Generate training loader and test loader.
for $i = 1$ to e **do**
    Extract spatial–spectral features $\boldsymbol{X}_H^{conv}$ from $\boldsymbol{X}_H^{pca}$, and extract elevation features $\boldsymbol{X}_L^{conv}$ from $\boldsymbol{X}_L$.
    Flatten 2D feature maps into 1D feature vectors.
    Perform tokenization transform with feature vectors to generate feature tokens by Eq. (1).
    Input the feature tokens to the transformer encoder for feature learning.
    Perform the transformer encoder by Eq. (2) and (3).
    Input the learned feature tokens to the CTA fusion encoder for feature fusion.
    Perform the CTA fusion encoder by Eq. (4), (5), and (6).
    Input the first *cls* tokens of the output of CTA to the MLP layer.
    Use a softmax function to identify the labels.
end for
Use the test set with the trained model to get predicted labels.

III. Experiments and Analysis

Four publicly accessible multimodal RS datasets were used for experiments to evaluate the effectiveness of the proposed framework. First, detailed information on the four datasets used for experiments is presented. Then, the details of the experimental settings are described, and ablation experiments are performed to demonstrate the functionality of the different modules in the proposed framework. Finally, experimental results are presented that demonstrate the effectiveness of the proposed method and its superiority over existing approaches.

A. Data Descriptions

To evaluate the effectiveness of the proposed framework, four datasets comprising both HSI and LiDAR data were selected for experiments: the Houston2013, Trento, Muufl Gulfport Hyperspectral and LiDAR Data (MUUFL) [56], [57], and Augsburg datasets.

1) Houston2013 Dataset: The Houston2013 dataset was provided by the 2013 IEEE GRSS Data Fusion Contest.[1] The dataset covers the University of Houston campus and the surrounding urban area and was collected by the NSF-funded National Center for Airborne Laser Mapping in 2012. In the Houston2013 dataset, the HSI data contains 144 spectral bands with a spectral resolution ranging from 0.38 to 1.05 μm. For the same region, LiDAR data are provided with only one band. Both the HSI and LiDAR data are made up of 349×1905 pixels with a spatial resolution of 2.5 m. This dataset depicts 15 urban land-cover classes. Fig. 5 visualizes a pseudo-color composite image of the HSI data, a grayscale image of the LiDAR data, and a ground-truth map.

2) Trento Dataset: The Trento dataset was captured over a rural region in the south of Trento, Italy. The HSI data were obtained by the AISA Eagle system for hyperspectral imaging (AISA) Eagle sensor with 63 spectral bands and a spectral

[1] http://www.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

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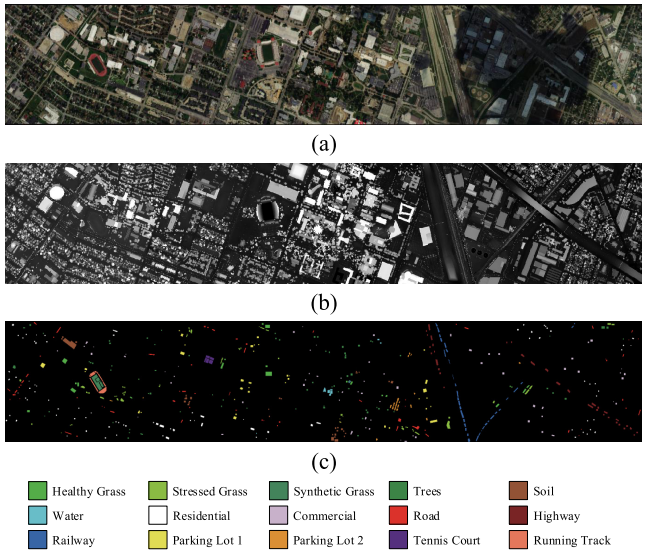


Fig. 5. Houston dataset. (a) Pseudo-color composite image based on bands 64, 43, and 20 for HSIs. (b) Grayscale image for LiDAR-based DSM. (c) Ground-truth map.

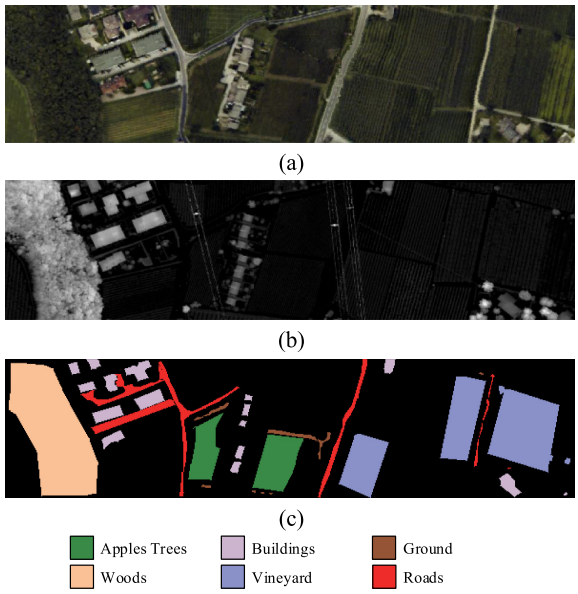


Fig. 6. Trento dataset. (a) Pseudo-color composite image based on bands 20, 15, and 5 for HSIs. (b) Grayscale image for LiDAR-based DSM. (c) Ground-truth map.

resolution ranging from 0.42 to 0.99 μm, and the LiDAR data were collected by the Optech ALTM 3100EA sensor with 1 raster. The size of this dataset is 600 $\times$ 166 pixels with a spatial resolution of 1 m and includes six different types of ground objects. Fig. 6(a)–(c) shows a pseudo-color composite image of the HSI data, a grayscale image of the LiDAR data, and a ground-truth map. respectively.

3) MUUFL Dataset: The MUUFL dataset was obtained over the area of the campus of the Gulf Park University of Southern Mississippi by the reflective optics system imaging spectrometer sensor in November 2010.[2] In the MUUFL dataset, the HSI data comprise 72 spectral bands ranging from

[2]https://github.com/GatorSense/MUUFLGulfport/

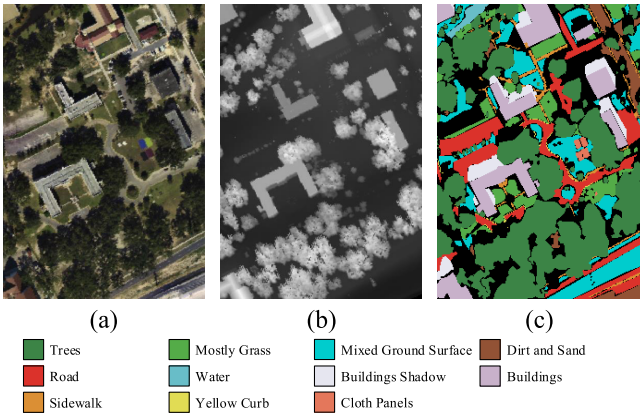


Fig. 7. MUUFL dataset. (a) Pseudo-color composite image based on bands 20, 15, and 5 for HSIs. (b) Grayscale image for the LiDAR-based DSM. (c) Ground-truth map.

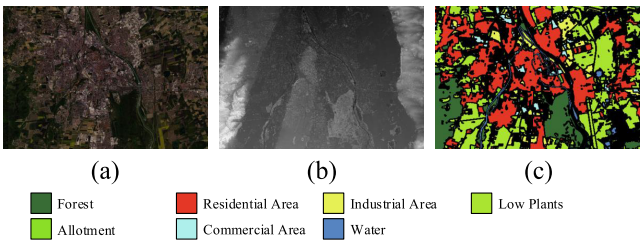


Fig. 8. Augsburg dataset. (a) Pseudo-color composite image based on bands 20, 15, and 10 of HSIs. (b) Grayscale image for the LiDAR-based DSM. (c) Ground-truth map.

0.38 to 1.05 μm, while the LiDAR data contain two rasters with a 1.06 μm wavelength. The first and last eight spectral bands were removed due to excessive noise. This dataset is made up of 325 $\times$ 220 pixels, totaling 11 different land-cover classes. A pseudo-color composite image of the HSI data, a grayscale image of the LiDAR data, and a ground-truth map are shown in Fig. 7(a)–(c), respectively.

4) Augsburg Dataset: The Augsburg dataset was captured over the city of Augsburg, Germany. The HSI data were acquired by the DAS-EOC HySpex sensor [58], and the LiDAR-based DSM data were collected by the DLR-3 K system [59]. The spatial resolutions of both images were down-sampled to a unified resolution of 30 m to adequately manage the multimodal fusion. In this dataset, the HSI data consist of 180 bands ranging from 0.4 to 2.5 μm, while the DSM data have a single raster. The size of this dataset is 332 $\times$ 485 pixels. The dataset depicts seven different land-cover classes. Fig. 8(a)–(c) shows the pseudo-color composite image of the HSI data, a grayscale image of the LiDAR data, and a ground-truth map, respectively.

The names of the land-cover categories, the number of training samples, and the number of test samples for these four datasets are listed in Table I.

B. Experimental Settings

1) Evaluation Indicators: To evaluate the classification performance of the proposed framework and other existing

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TABLE I

TRAINING AND TEST SAMPLE NUMBERS FOR HOUSTON, TRENTO, MUUFL, AND AUGSBURG DATASETS

NO.	Houston2013 dataset			Trento dataset			MUUFL dataset			Augsburg dataset		
	Class Name	Training.	Test.	Class Name	Training.	Test.	Class Name	Training.	Test.	Class Name	Training.	Test.
C01	Healthy Grass	198	1053	Apple Trees	129	3905	Trees	150	23096	Forest	675	12832
C02	Stressed Grass	190	1064	Buildings	125	2778	Mostly Grass	150	4120	Residential Area	1516	28813
C03	Synthetic Grass	192	505	Ground	105	374	Mixed Ground Surface	150	6732	Industrial Area	192	3659
C04	Trees	188	1056	Woods	154	8969	Dirt and Sand	150	1676	Low Plants	1342	25515
C05	Soil	186	1056	Vineyard	184	10317	Road	150	6537	Allotment	28	547
C06	Water	182	143	Roads	122	3052	Water	150	316	Commercial Area	82	1563
C07	Residential	196	1072				Buildings Shadow	150	2083	Water	76	1454
C08	Commercial	191	1053				Buildings	150	6090			
C09	Road	193	1059				Sidewalk	150	1235			
C10	Highway	191	1036				Yellow Curb	150	33			
C11	Railway	181	1054				Cloth Panels	150	119			
C12	Parking Lot1	192	1041									
C13	Parking Lot2	184	285									
C14	Tennis Court	181	247									
C15	Running Track	187	473									
-	Total	2832	12197	Total	819	29395	Total	1650	52037	Total	3911	74383

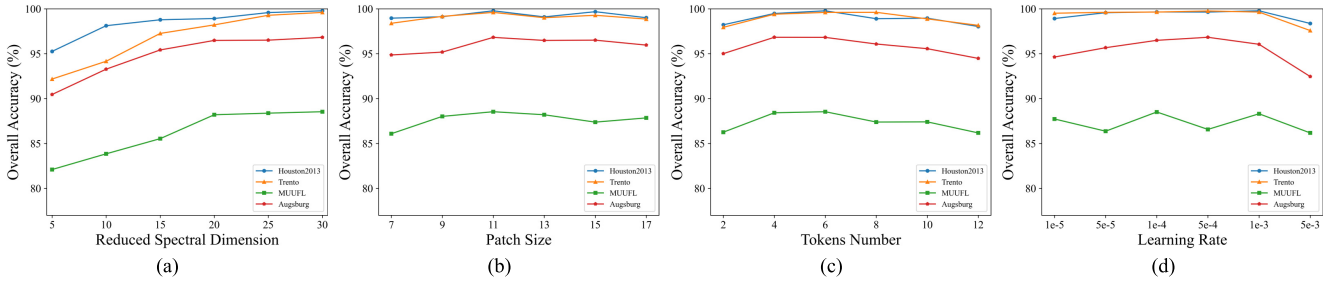


Fig. 9. Influence of different parameters on the overall accuracy. (a) Reduced spectral dimension. (b) Patch size. (c) Tokens number. (d) Learning rate.

models, four widely used quantity analysis criteria were computed, namely the overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and per-class accuracy. For each indicator, a higher value indicates a more accurate classification.

2) Configurations: The experiments with the proposed method and other deep-learning methods were all implemented in the PyTorch platform, using an AMAX tower server with an Inter Xeon Silver 4210 2.20 GHz CPU, 128 GB RAM, and an NVIDIA GeForce RTX 2080Ti GPU with 11 GB of VRAM. To optimize the network, the Adam optimizer was selected as the initial optimizer. For the training stage, the mini-batch size, and the number of training epochs were set to 64, and 100, respectively.

C. Parameter Analysis

Several hyperparameters that may affect the classification performance and training process were analyzed, including the reduced spectral dimension b, the patch size s of input data patches, the tokens number ω_h and ω_l, the learning rate, and the depth of encoder L.

1) Reduced Spectral Dimension: To evaluate the effect of the reduced spectral dimension b, the other hyperparameter values were fixed and b was selected from a candidate set $\{1, 5, 10, 15, 20, 25, 30\}$. Fig. 9(a) shows the OA of the proposed framework in four datasets by varying the reduced spectral dimensions. It can be seen that as b increases, OA initially increases and then tends toward a stable state. Considering the computational complexity and classification performance, we concluded that b could be set to 30.

2) Patch Size: Similar to the analysis of b, the other hyperparameters were fixed. We select the patch sizes s from a candidate set $\{7, 9, 11, 13, 15, 17\}$ to evaluate its effect. Fig. 9(b) shows the effect of different patch sizes s on the OA. When the patch size was set to 11, the optimal OA values were achieved in four datasets.

3) Tokens Number: Fig. 9(c) shows the influence of the tokens number on the OA of the four datasets. ω_h and ω_l were set to equal values for the parameter experimental analysis. Token number was selected from a candidate set $\{2, 4, 6, 8, 10, 12\}$. It can be seen that the best OA was achieved with the token number of 4 or 6 in four datasets.

4) Learning Rate: The learning rate, an important hyperparameter in the deep-learning model, significantly impacts the objective function convergence to the local optimal value. The objective function can quickly reach its local minimum with an appropriate learning rate. In experiments, the learning rate was selected from a candidate set {1e–5, 5e–5, 1e–4, 5e–4, 1e–3, 5e–3}. Fig. 9(d) shows the OA of the proposed method in four datasets by setting different learning rates. It can be observed that the optimal learning rate for the Houston2013 and Trento datasets is 1e–3, while for the MUUFL dataset, it is 1e–4, and for the Augsburg dataset, it is 5e–4.

5) Depth of the Encoder: As shown in Fig. 1, we consider the TE and CTA fusion encoder as a unified encoder structure.

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TABLE II
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE HOUSTON2013 DATASET (OPTIMAL RESULTS ARE BOLDED)

NO.	Only HSI Input					HSI and LiDAR Input				
	SVM [21]	RF [23]	CNN2D [36]	CNN3D [38]	SSFTT [52]	S2FL [32]	EndNet [42]	CCRNet [45]	CoupledCNN [44]	Proposed
C01	92.66±07.25	62.50±01.84	95.21±03.12	96.47±01.91	99.10±00.93	97.82±00.84	96.13±00.80	93.92±04.27	93.64±03.77	**99.48±00.51**
C02	89.89±06.23	81.57±01.43	96.83±03.71	98.93±01.13	99.12±00.85	87.11±02.38	94.72±00.87	94.46±05.22	96.86±03.89	**99.84±00.15**
C03	98.69±00.42	83.19±01.40	91.78±07.35	98.96±00.85	**99.91±00.09**	99.02±00.41	98.73±00.72	99.82±00.25	99.39±00.55	99.84±00.16
C04	91.58±03.19	85.89±01.40	93.78±02.27	99.20±00.68	99.01±00.60	92.20±03.34	91.50±03.39	99.52±00.41	97.37±02.75	**99.74±00.24**
C05	98.89±00.38	78.61±02.75	98.45±01.62	**100.00±00.00**	99.98±00.06	98.50±00.52	98.98±00.38	99.69±00.30	99.30±00.60	**100.00±00.00**
C06	85.59±02.17	79.08±02.48	63.55±11.99	92.01±03.86	**100.00±00.00**	84.83±02.68	93.22±00.87	99.44±01.13	90.25±07.58	99.99±00.03
C07	79.24±01.05	52.63±02.93	89.22±04.12	93.44±03.92	98.30±01.20	83.18±02.82	89.25±01.12	96.57±01.38	95.89±02.60	**99.64±00.57**
C08	49.76±03.30	91.19±03.20	79.19±03.84	91.71±03.85	98.34±00.90	53.25±03.33	76.52±02.06	92.39±05.22	92.49±04.94	**99.43±00.79**
C09	80.42±01.21	50.12±04.23	81.02±03.48	89.21±04.76	98.90±00.95	80.52±01.24	86.05±01.09	90.66±05.49	90.40±03.89	**99.88±00.27**
C10	51.78±03.02	81.32±02.78	88.02±02.93	96.96±02.68	99.95±00.14	73.11±01.89	91.28±01.12	92.61±03.75	96.53±02.17	**100.00±00.00**
C11	62.53±01.31	86.91±03.04	90.63±02.14	96.75±01.13	99.75±00.31	64.75±02.85	87.60±01.83	95.27±03.84	98.28±01.27	**99.98±00.04**
C12	61.45±08.60	78.43±05.08	89.14±03.71	96.25±02.69	99.25±00.62	69.24±03.17	83.18±01.31	91.28±05.26	91.01±05.68	**99.66±00.35**
C13	34.88±05.28	96.35±04.14	69.10±12.50	94.19±02.62	99.67±00.38	43.51±01.96	84.18±02.27	96.81±02.41	92.04±06.69	**99.72±00.68**
C14	98.18±00.72	93.91±02.50	76.83±07.00	**100.00±00.00**	**100.00±00.00**	98.95±00.47	98.33±00.60	**100.00±00.00**	99.74±00.38	**100.00±00.00**
C15	98.99±00.38	77.36±01.29	96.65±02.96	**100.00±00.00**	**100.00±00.00**	99.18±00.41	98.99±00.38	99.75±00.34	99.89±00.25	**100.00±00.00**
OA(%)	77.53±00.63	83.69±00.53	88.89±01.35	96.19±00.72	99.30±00.13	81.26±00.39	90.18±00.87	95.28±00.70	95.48±00.88	**99.78±00.11**
AA(%)	78.44±00.38	76.53±00.54	86.61±01.98	96.27±00.63	99.42±00.13	81.78±00.37	90.28±00.25	96.15±00.55	95.51±01.11	**99.81±00.09**
$\kappa\times100$	75.61±00.68	81.40±00.57	88.16±01.49	95.88±00.78	99.23±00.13	79.66±00.43	88.72±00.34	94.87±00.76	95.11±00.95	**99.77±00.12**

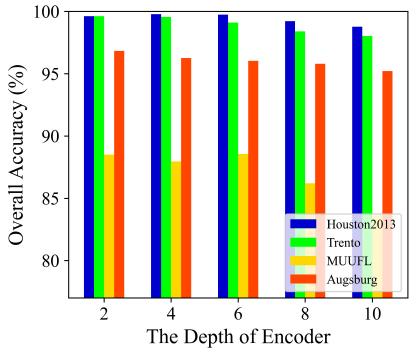


Fig. 10. OA using different encoder depths on four datasets.

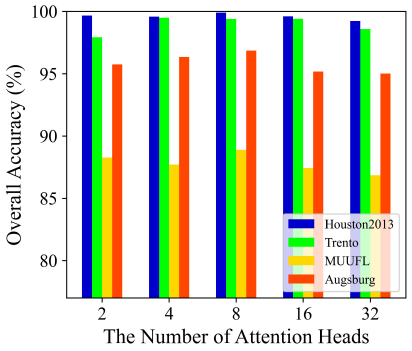


Fig. 11. OA using different numbers of attention heads on four datasets.

The depth of encoder L can affect the feature representation capability. To verify the influence of L on classification performance, the candidate set was set as $\{2, 4, 6, 8, 10\}$. The OAs with different depths of the encoder in four datasets are shown in Fig. 10. It can be seen that when the depth of the encoder is 2, the proposed network can achieve the optimal classification effect. As the depth of the encoder increases, the performance of the network on four datasets has a downward trend. This phenomenon illustrates that a deeper network does not lead to better performance.

6) Attention Heads: The MHSA module uses multiple attention heads to learn the correspondence of different representation subspaces, in which each head corresponds to an independent subspace with regard to a different feature representation. Therefore, different numbers of attention heads can affect the final feature representation capability of the transformer, thereby affecting the classification performance. Fig. 11 shows the OAs with different numbers of attention heads on four datasets, and the candidate set of attention heads is $\{2, 4, 8, 16, 32\}$. The experimental results show that the reasonable numbers of attention heads for four datasets are 8, 4, 8, and 8, respectively.

D. Classification Results and Analysis

To verify the effectiveness of the proposed framework, experiments were performed to compare it against several representative classification methods, including SVM [21], RF [23], 2-D-CNN [36], 3-D-CNN [38], the recently proposed SSFTT [52], shared and specific feature learning (S2FL) [32], EndNet [42], cross-channel reconstruction network (CCRNet) [45], and CoupledCNN [44], where SVM, RF, 2-D-CNN, 3-D-CNN, and SSFTT were designed for HSI classification tasks. For these methods, the settings of networks were set as described in their corresponding references. As shown in Table I, the training sample sets and test sample sets for the above methods were chosen at random to provide a fair comparison.

1) Quantitative Results and Analysis: Tables II–V clearly show the OA, AA, κ, and per-class accuracy of the proposed method and each comparative method for the Houston2013, Trento, MUUFL, and Augsburg datasets, respectively. In each table, the best results are highlighted in bold. The values of the evaluation indicators show that the proposed framework performed the best and was often significantly more accurate than other methods. Taking Houston2013 as an example,

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TABLE III

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE TRENTO DATASET (OPTIMAL RESULTS ARE BOLDED)

NO.	Only HSI Input					HSI and LiDAR Input				
	SVM [21]	RF [23]	CNN2D [36]	CNN3D [36]	SSFTT [52]	S2FL [32]	EndNet [42]	CCRNet [45]	CoupledCNN [44]	Proposed
C01	79.55±01.44	86.22±03.55	98.26±01.01	99.37±00.62	98.53±00.41	80.53±01.91	87.50±00.70	99.26±01.21	**99.54±00.30**	99.10±00.43
C02	76.48±04.69	85.10±03.01	89.99±02.87	91.66±02.04	98.06±01.50	79.86±03.55	87.36±00.91	96.79±01.84	97.61±01.98	**98.95±00.41**
C03	85.64±02.38	98.42±00.57	97.27±01.87	99.10±00.99	**100.00±00.00**	90.21±01.71	98.23±00.91	94.08±02.27	98.34±00.19	**100.00±00.00**
C04	92.26±02.18	93.62±00.62	99.79±00.36	99.98±00.04	99.99±00.02	93.04±00.71	98.34±00.19	99.98±00.03	**100.00±00.00**	**100.00±00.00**
C05	82.10±01.33	85.20±01.80	99.15±00.65	99.45±00.92	99.97±00.04	82.58±01.26	93.68±00.26	99.96±00.06	**100.00±00.00**	99.99±00.01
C06	81.85±02.53	74.26±02.73	87.55±02.52	89.89±03.21	97.40±01.31	80.32±02.49	86.59±01.29	96.39±01.82	97.69±01.44	**98.45±01.22**
OA(%)	84.35±00.69	86.93±00.25	96.61±00.55	97.80±00.26	99.34±00.17	85.11±00.56	92.92±00.12	98.53±00.47	98.79±00.38	**99.60±00.11**
AA(%)	82.98±00.54	87.14±00.63	95.33±00.57	96.46±00.39	98.95±00.30	84.43±00.62	91.78±00.21	95.52±01.33	97.70±00.37	**99.34±00.17**
$\kappa\times100$	79.21±00.90	82.59±00.34	96.13±00.40	97.05±00.35	99.11±00.22	80.21±00.72	90.53±00.16	98.03±00.63	98.38±00.50	**99.46±00.15**

TABLE IV

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE MUUFL DATASET (OPTIMAL RESULTS ARE BOLDED)

NO.	Only HSI Input					HSI and LiDAR Input				
	SVM [21]	RF [23]	CNN2D [36]	CNN3D [38]	SSFTT [52]	S2FL [32]	EndNet [42]	CCRNet [45]	CoupledCNN [44]	Proposed
C01	74.24±02.05	81.27±02.43	86.59±04.63	84.78±02.97	87.28±01.86	80.95±00.76	84.16±00.86	84.67±01.57	87.15±03.62	**89.63±00.93**
C02	72.79±02.42	76.23±03.92	81.64±07.93	82.92±03.51	82.78±02.21	76.95±02.51	83.23±01.84	85.91±01.46	**87.20±01.46**	86.04±02.25
C03	57.02±02.77	61.26±02.89	68.31±09.67	72.99±04.77	79.21±01.29	66.06±02.47	71.77±02.52	66.07±04.01	79.96±06.29	**82.53±01.90**
C04	62.37±02.16	74.81±02.24	92.12±02.99	95.37±01.92	94.56±01.81	78.68±02.97	87.70±01.36	94.68±02.49	94.71±03.35	**95.61±01.58**
C05	78.76±02.45	83.40±01.10	87.03±02.55	83.72±05.30	82.15±01.53	84.16±01.84	88.86±01.53	85.82±04.01	**89.76±02.65**	84.29±01.82
C06	91.71±01.58	90.63±01.19	98.67±01.24	**99.84±00.31**	99.62±00.49	94.46±00.72	94.34±01.35	98.97±00.53	99.05±00.60	99.65±00.51
C07	82.34±01.88	82.48±02.00	92.70±02.25	90.36±02.63	93.34±02.45	84.18±01.51	88.69±01.43	88.56±02.95	91.54±02.35	**94.59±00.94**
C08	65.28±01.86	73.11±01.90	87.88±02.43	93.08±03.53	91.35±02.91	76.39±01.88	80.44±01.77	94.79±00.97	**96.88±01.43**	94.51±00.84
C09	**84.05±02.09**	59.47±01.70	76.39±02.97	70.56±02.67	70.79±05.34	71.64±03.04	75.36±02.85	65.25±03.57	70.98±03.03	78.84±02.52
C10	81.51±04.39	80.61±09.81	94.55±05.30	93.03±04.05	92.42±05.20	92.42±04.10	97.27±02.23	87.87±06.06	94.18±05.74	**98.47±01.11**
C11	74.20±05.06	94.71±02.66	98.74±01.44	99.50±00.59	98.57±01.73	85.88±02.65	98.15±00.95	97.98±01.47	98.46±00.72	**99.89±00.18**
OA(%)	71.70±01.08	77.04±00.65	83.93±02.16	84.20±01.31	86.62±00.79	78.51±00.43	82.84±00.39	83.12±00.92	88.43±01.53	**88.54±00.67**
AA(%)	74.93±01.06	78.00±00.85	86.97±01.02	87.83±00.69	88.37±00.89	82.07±03.14	86.34±00.26	86.08±00.69	90.34±02.19	**91.21±00.50**
$\kappa\times100$	64.65±01.20	70.49±00.72	79.73±02.86	79.77±01.63	82.80±00.78	72.50±00.54	77.72±00.77	77.95±01.38	85.07±01.61	**85.22±00.84**

TABLE V

CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE AUGSBURG DATASET (OPTIMAL RESULTS ARE BOLDED)

NO.	Only HSI Input					HSI and LiDAR Input				
	SVM [21]	RF [23]	CNN2D [36]	CNN3D [38]	SSFTT [52]	S2FL [32]	EndNet [42]	CCRNet [45]	CoupledCNN [44]	Proposed
C01	95.35±00.40	98.09±00.32	93.97±01.54	97.66±00.53	97.72±00.54	97.20±00.29	94.50±00.55	96.56±00.65	**98.66±00.74**	98.11±00.30
C02	89.97±00.36	78.21±01.60	95.94±02.57	97.61±01.61	94.42±01.36	72.31±01.20	88.63±00.55	96.62±00.54	98.55±00.71	**98.60±00.42**
C03	06.50±01.54	29.89±03.89	66.13±13.45	78.17±05.16	82.85±03.29	32.39±03.16	41.34±03.25	82.77±04.61	84.46±05.60	**90.25±02.34**
C04	66.75±01.45	80.17±01.81	94.49±01.79	98.32±00.38	96.48±00.98	87.38±01.03	94.21±00.39	98.00±00.13	**98.64±00.46**	98.42±00.32
C05	06.58±02.64	08.36±01.69	06.43±02.97	24.51±08.75	**81.94±05.43**	40.63±05.43	31.72±02.95	41.34±07.96	68.16±05.19	79.71±05.35
C06	10.82±01.20	42.88±03.94	16.58±04.67	22.67±06.73	**80.92±03.80**	39.96±02.95	28.36±04.73	33.29±05.79	72.12±04.49	74.78±03.52
C07	51.64±02.21	63.93±01.13	47.96±04.57	52.23±05.17	66.97±01.77	**70.33±01.85**	52.63±01.80	58.41±03.62	62.68±06.44	68.57±02.85
OA(%)	75.78±00.58	78.27±00.67	90.38±00.94	93.76±00.50	94.64±00.53	78.74±00.38	86.62±00.23	94.62±00.49	95.38±00.56	**96.82±00.22**
AA(%)	46.80±00.62	57.19±00.69	60.17±02.06	66.62±01.83	85.73±01.65	62.88±00.81	59.93±00.63	73.01±03.17	82.37±02.89	**86.92±00.89**
$\kappa\times100$	64.21±00.82	69.41±00.88	86.09±01.34	90.98±00.71	92.34±00.78	70.54±00.48	80.71±00.32	93.39±00.41	93.63±00.43	**95.74±00.96**

as shown in Table II, the OA value of the proposed framework was 99.78%, which is competitive in the HSI and LiDAR joint classification task. For classes with similar object materials, such as two different parking lots, the proposed method achieved a significant improvement of 8.38% and 2.91%. This is because of the use of the dual-branch feature learning stage and a CTA fusion strategy. As can be seen in Table IV, the result of C05 and C09 are not well among all classification methods. C05 represents the "road," while C09 represents the "sidewalk." It can be seen from the ground-truth map that the sample distributions of the two categories are mostly adjacent and relatively dispersed. The reason for the poor classification performance of the two categories is that the spectral reflection curves are very similar due to the same land cover material of the categories, and the elevations are also similar, resulting in many misclassified samples between the two categories.

As shown in Fig. 12, to measure the stability and robustness of the proposed method with different percentages of training samples, 5%, 10%, 15%, and 20% labeled samples were randomly selected as training data for the Houston2013 and

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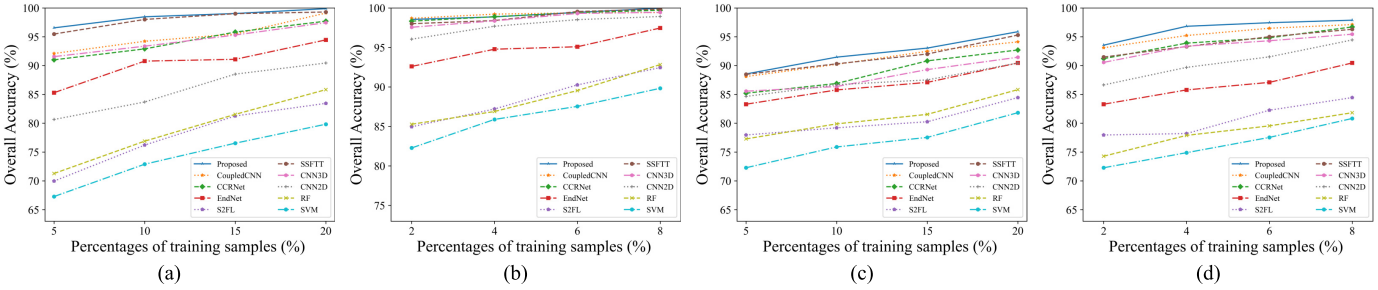


Fig. 12. OA of different models with different training samples percentages. (a) Houston2013 dataset. (b) Trento dataset. (c) MUUFL dataset. (d) Augsburg dataset.

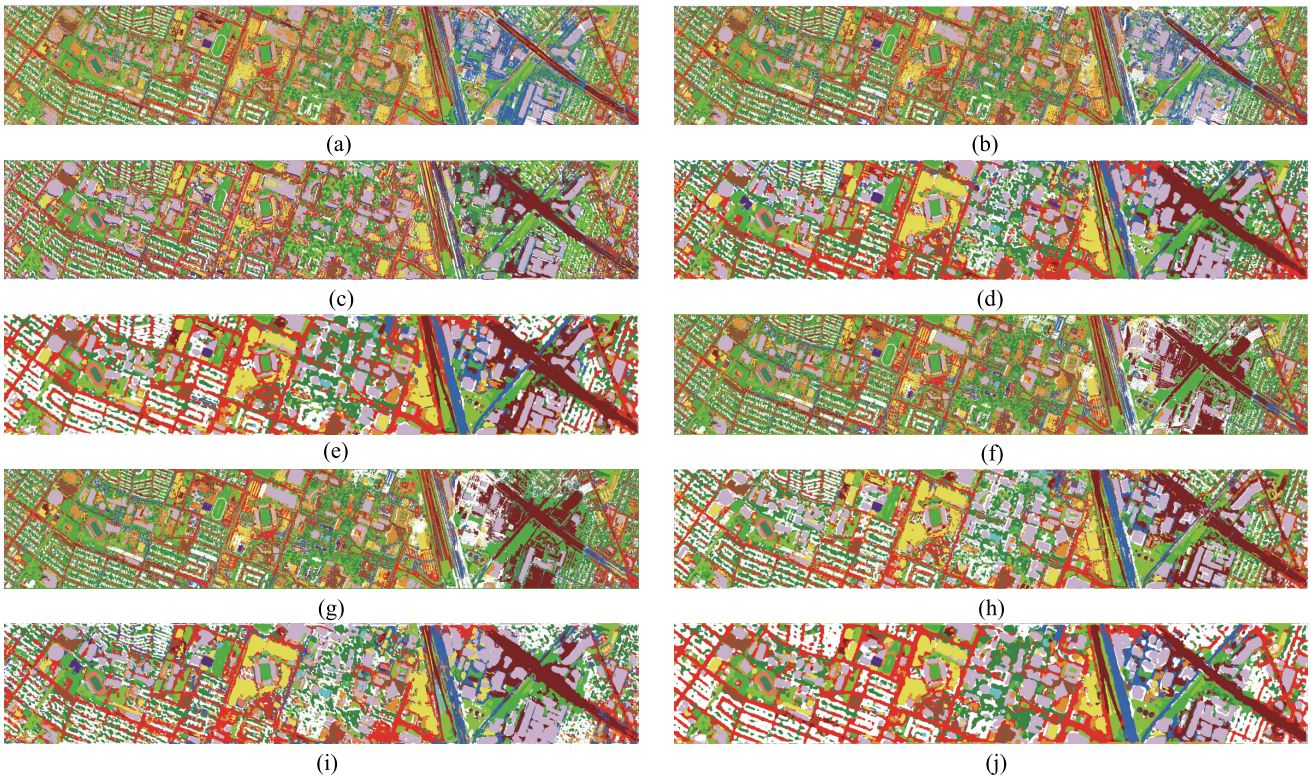


Fig. 13. Classification maps by different methods on the Houston2013 dataset. (a) SVM (77.53%). (b) RF (83.69%). (c) CNN2D (88.89%). (d) CNN3D (96.19%). (e) SSFTT (99.30%). (f) S2FL (81.26%). (g) EndNet (90.18%). (h) CCRNet (95.28%). (i) CoupledCNN (95.48%). (j) Proposed (99.78%).

MUUFL datasets, and 2%, 4%, 6%, and 8% labeled samples were randomly selected as training data for the Trento and Augsburg datasets. With a small number of samples, the proposed method still performed well. In addition, when the sample proportion was 10%, all the other methods were less accurate for the Houston2013 dataset. As the number of samples increased, the OA did not indicate a significant difference for the Houston2013 and Trento datasets because the accuracy was almost 100% even for the small samples. From the experiments on the four datasets, it can be seen that the proposed method maintained the best classification performance over the whole range of sample sizes.

2) Visual Evaluation and Analysis: The classification maps of several comparison methods are shown in Figs. 13–16 for the four datasets, respectively. It can be seen from the maps that the proposed method produces a better classification performance with clearer boundaries when compared with the other methods, which is consistent with the numerical results.

TABLE VI

TRAINING TIME IN MINUTES (m) AND TEST TIME IN SECONDS (s) BETWEEN THE CONTRAST METHODS AND PROPOSED METHOD ON THREE DATASETS (OPTIMAL RESULTS ARE BOLDED AND SUBOPTIMAL RESULTS ARE UNDERLINED)

Methods	Houston2013		Trento		MUUFL		Augsburg	
	Train.(m)	Test (s)	Train.(m)	Test (s)	Train.(m)	Test (s)	Train.(m)	Test (s)
CNN2D [36]	1.68	3.57	**2.12**	**4.68**	1.89	4.34	2.61	15.42
CNN3D [38]	7.13	4.90	6.29	9.62	8.29	10.09	11.69	28.37
SSFTT [52]	6.67	4.50	3.02	5.87	7.24	8.05	2.81	13.61
EndNet [42]	**1.54**	**2.98**	2.58	4.91	**1.73**	**4.18**	**1.48**	**12.47**
CCRNet [45]	8.77	3.63	6.32	7.40	8.54	8.18	13.25	25.89
CoupledCNN [44]	3.04	3.83	2.88	5.86	2.61	7.86	6.46	16.48
Proposed	7.25	4.91	5.32	6.43	7.73	8.16	15.88	39.79

The first few pixel-level methods of SVM, RF, and S2FL contained significant noise for the four datasets. These models cannot accurately identify the types of objects. For the Houston2013 dataset, by comparing some of the boundary regions shown in Fig. 13, it is concluded that our method produces the

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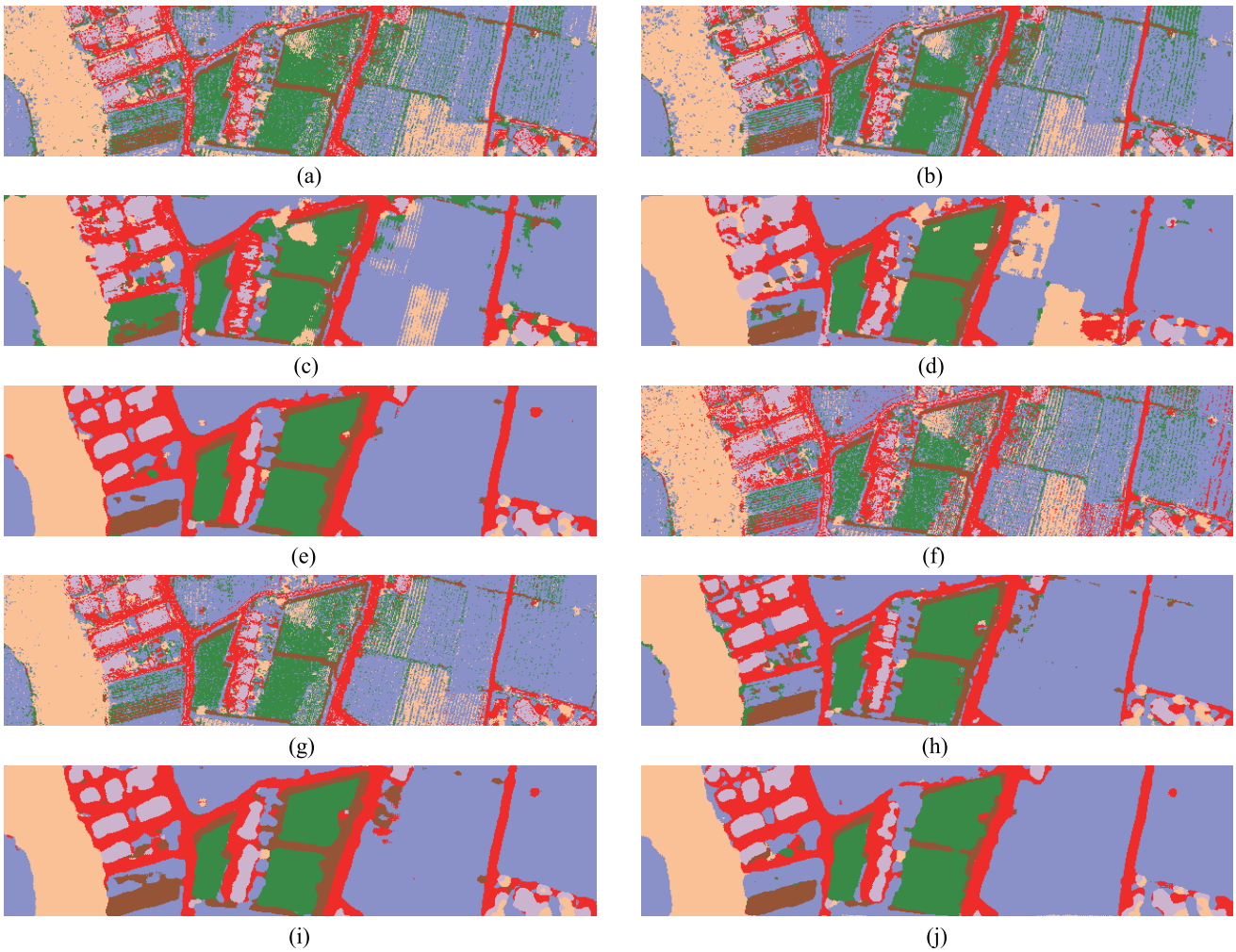


Fig. 14. Classification maps by different methods on the Trento dataset. (a) SVM (84.35%). (b) RF (86.93%). (c) CNN2D (96.61%). (d) CNN3D (97.80%). (e) SSFTT (99.34%). (f) S2FL (85.11%). (g) EndNet (92.92%). (h) CCRNet (98.53%). (i) CoupledCNN (98.79%). (j) Proposed (99.60%).

TABLE VII
ABLATION ANALYSIS OF DIFFERENT MODAL DATA INPUTS (OPTIMAL RESULTS ARE BOLDED)

Cases	Houston2013			Trento			MUUFL			Augsburg		
	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$
Only HSI	99.12	99.13	98.93	99.32	99.19	99.11	86.86	88.59	84.81	94.68	85.46	92.26
Only LiDAR	64.52	63.47	62.58	91.08	86.57	87.96	47.35	45.29	45.68	78.36	75.58	77.35
HSI+LiDAR	**99.78**	**99.81**	**99.77**	**99.60**	**99.34**	**99.46**	**88.54**	**91.21**	**85.22**	**96.82**	**86.92**	**95.74**

most accurate boundary regions, which further demonstrates the classification performance of the proposed method.

3) Computation Time Cost Comparison: A comparison of training and testing time for CNN2-D [36], CNN3-D [38], SSFTT [52], EndNet [42], CCRNet [45], and Coupled-CNN [44] and the proposed method is shown in Table VI. Due to the limited fusion efficiency of the CTA fusion encoder in the feature fusion stage, the proposed method is more computationally time-consuming than the other methods.

E. Ablation Analysis

1) Ablation Analysis of Different Modal Data Inputs: Considering the influence of different modal data on the classification performance of the model, we conducted two sets of experiments, single HSI data input and single LiDAR data input, respectively. The experimental results are shown in Table VII. By comparing OA and AA values on four datasets, the proposed fusion network achieves better classification results. This validates that the two kinds of data play a positive role in improving the classification performance and also verifies that the designed dual-branch network can make full use of the effective information of different modal data.

2) Ablation Analysis of Different Components: Since the proposed network benefits from many components, we analyzed the contribution of each component to the joint classification performance by ablation experiments. Specifically, the proposed framework was mainly divided into five components—Conv3-D for HSIs, Conv2-D for HSIs and

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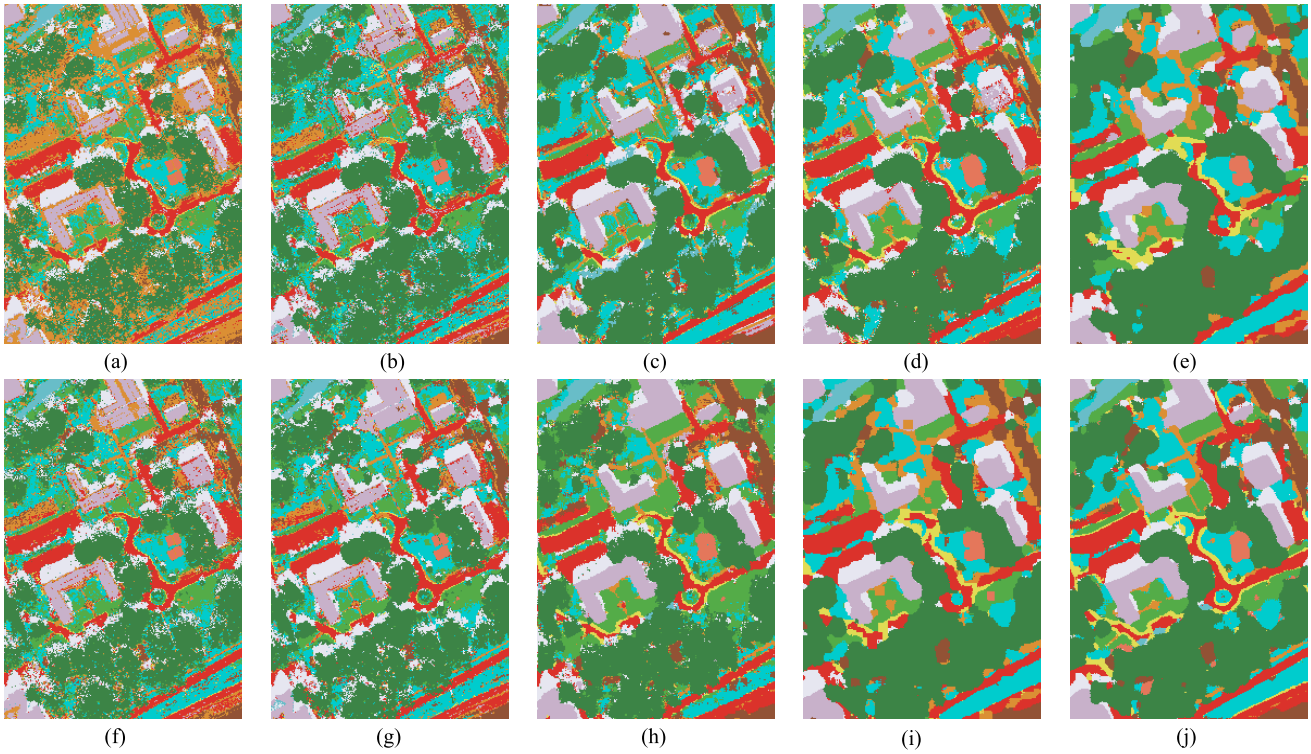


Fig. 15. Classification maps by different methods on the MUUFL dataset. (a) SVM (71.70%). (b) RF (77.04%). (c) CNN2D (83.93%). (d) CNN3D (84.20%). (e) SSFTT (86.62%). (f) S2FL (78.51%). (g) EndNet (82.84%). (h) CCRNet (83.12%). (i) CoupledCNN (88.43%). (j) Proposed (88.54%).

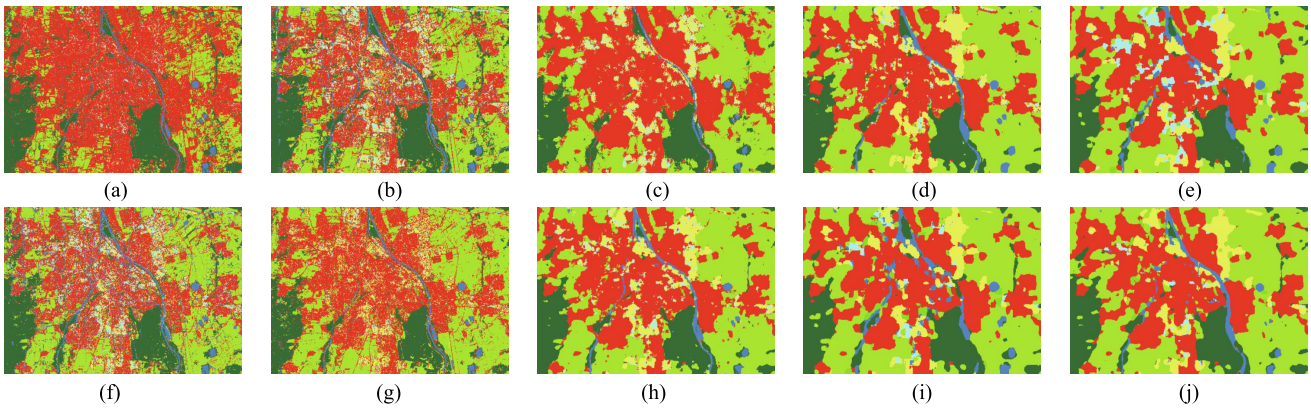


Fig. 16. Classification maps by different methods on the Augsburg dataset. (a) SVM (75.78%). (b) RF (78.27%). (c) CNN2D (90.38%). (d) CNN3D (93.76%). (e) SSFTT (94.64%). (f) S2FL (78.74%). (g) EndNet (86.62%). (h) CCRNet (94.62%). (i) CoupledCNN (95.38%). (j) Proposed (96.82%).

TABLE VIII

ABLATION ANALYSIS OF DIFFERENT COMPONENTS IN THE PROPOSED MODEL ON THE HOUSTON2013 DATASET (OPTIMAL RESULTS ARE BOLDED)

Cases	Component					Indicators		
	Conv3D	Conv2D	Tokenizer	TE	CTA	OA (%)	AA (%)	$\kappa \times 100$
1	√	-	√	√	√	82.46	68.32	79.59
2	-	√	√	√	√	93.43	91.02	92.62
3	-	-	PE	√	√	91.88	87.56	89.13
4	√	√	PE	√	√	96.52	95.43	95.89
5	√	√	-	-	-	94.62	92.26	91.48
6	√	√	√	√	-	99.30	99.19	99.27
7	√	√	√	√	√	**99.78**	**99.81**	**99.77**

LiDAR, tokenizers for HSIs and LiDAR, TE for HSIs and LiDAR, and CTA fusion encoder for HSIs and LiDAR.

As listed in Table VIII, the Houston2013 dataset was used to evaluate six variants of the proposed framework that had varying combinations of components removed. The classification accuracy of these variants was analyzed to evaluate the impact of the various components on the model. The first variant without Conv2-D components yielded the worst classification accuracy. The second variant without the Conv3-D component performed a little better than the first variant. The classification accuracy of the first two model variants verifies the salient effect of the convolutional module in the feature extraction stage. In the third variant, both Conv2-D and Conv3-D components were removed, however, the patch embedding (PE) method in Vision Transformer (ViT) [49] was introduced to use the TE component. The data after PCA dimension reduction were used directly in the PE operation. This variant obtained the OA of 91.88%. In the fourth variant, only the tokenizers component, which converts the HSI and

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LiDAR features, was replaced by PE—the OA value was 96.52%, below the optimal value. This shows that the tokenizers components contributed to the accuracy of the proposed model. The fifth variant contains only Conv2-D and Conv3-D components and has an OA of 94.62%, which is below the optimal value. This illustrates that the processing of features by the TE and CTA components contributed to the performance of the proposed framework. The sixth variant achieved the OA of 99.30% without the CTA fusion encoder component, slightly less than the optimal value. This shows that the fusion of feature tokens by the CTA fusion encoder contributed to the performance improvement. This analysis further confirms the validity of the proposed network framework, showing that all of its major components are important.

IV. Conclusion

In this article, a novel hierarchical CNN and transformer network was proposed for accurate joint classification of HSI and LiDAR data. Combining CNN-based feature exaction with transformer-based feature learning, a dual-branch network framework was designed for fully effective feature mining. An innovative CTA fusion encoder was created to fuse the two types of features during the fusion phase. Experiments on four groups of HSI and LiDAR data were performed to compare the method with existing classification methods, and the results verified that the proposed method achieved effective and superior performance. In conclusion, the proposed hierarchical CNN and transformer network can extract more discriminative information and effectively fuse heterogeneous data characteristics for HSI and LiDAR joint classification.

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